

# **LECTURE 3**

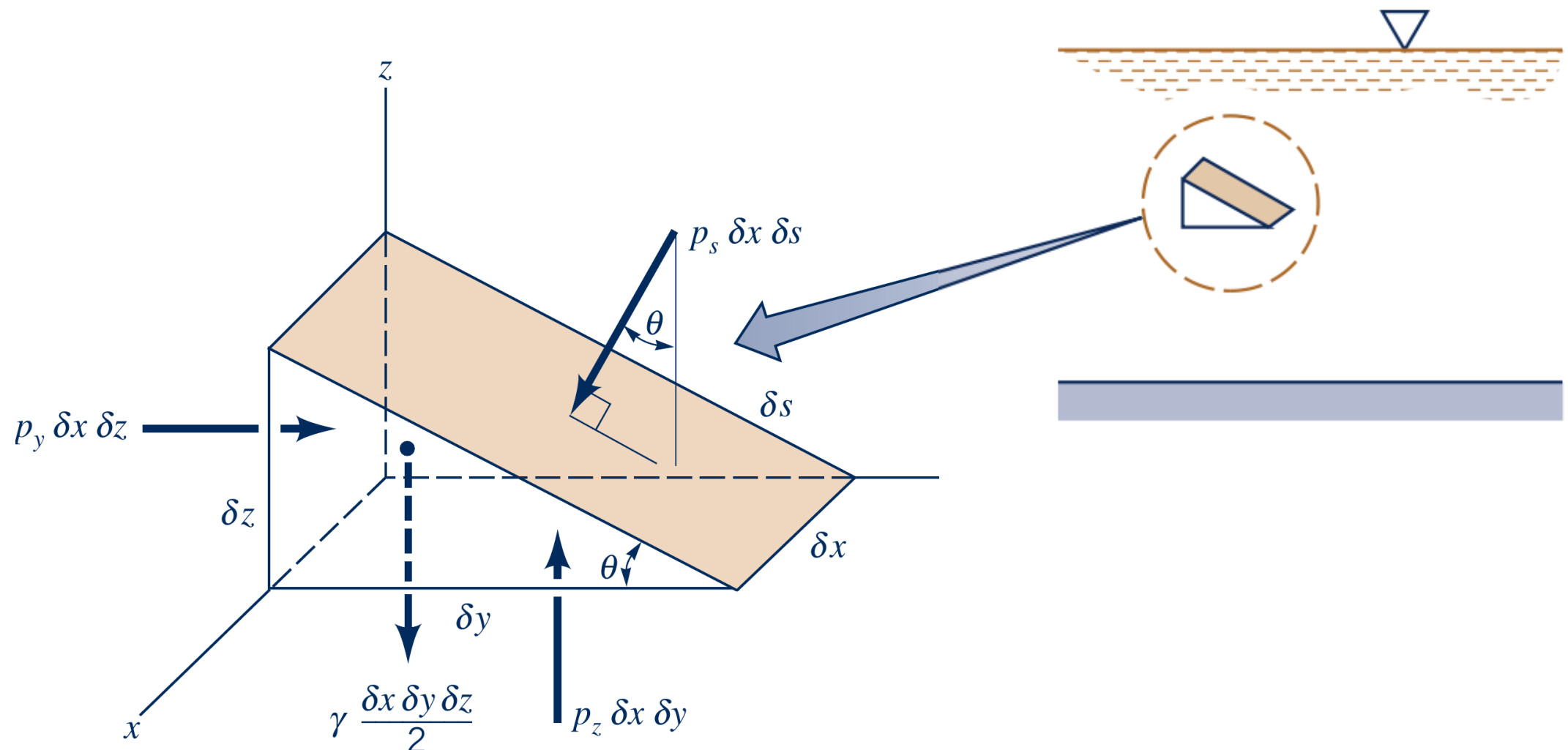
## **Fluid Statics: Pressure Variation and Measurement**

Reference: Munson, Rothmayer, Okiishi & Huebsch, Fundamentals of Fluid Mechanics, Chapter 2  
Wladimir E. Banda-Barragán

# Pressure at a point

**We assume there are no shearing stresses, so the only surface force is pressure:** the normal force per unit area on a plane inside the fluid.

Does its value at a point depend on the orientation of that plane?



*Forces on an arbitrary wedge-shaped element of fluid. The only external forces are pressure and weight.*

# The equations of motion for the wedge

Newton's second law,  $F = ma$ , in the y and z directions:

$$\sum F_y = p_y \delta x \delta z - p_s \delta x \delta s \sin \theta = \rho \frac{\delta x \delta y \delta z}{2} a_y$$

$$\sum F_z = p_z \delta x \delta y - p_s \delta x \delta s \cos \theta - \gamma \frac{\delta x \delta y \delta z}{2}$$

$$= \rho \frac{\delta x \delta y \delta z}{2} a_z$$

The weight of the wedge is  $\gamma \delta x \delta y \delta z / 2$ , and its mass is  $\rho \delta x \delta y \delta z / 2$ .

$$\delta y = \delta s \cos \theta, \quad \delta z = \delta s \sin \theta$$

## Taking the limit

Substituting  $\delta y = \delta s \cos \theta$  and  $\delta z = \delta s \sin \theta$ , and dividing through by  $\delta x \delta y \delta z$ :

$$p_y - p_s = \rho a_y \frac{\delta y}{2}$$

$$p_z - p_s = (\rho a_z + \gamma) \frac{\delta z}{2}$$

Now let the element shrink to a point, keeping the angle  $\theta$  fixed.

$$\delta x, \delta y, \delta z \rightarrow 0 : \quad p_y = p_s, \quad p_z = p_s$$

The three pressures are therefore equal:

$$p_s = p_y = p_z$$



# Pascal's law

The angle  $\theta$  was chosen arbitrarily, so the result holds for every orientation of the plane.

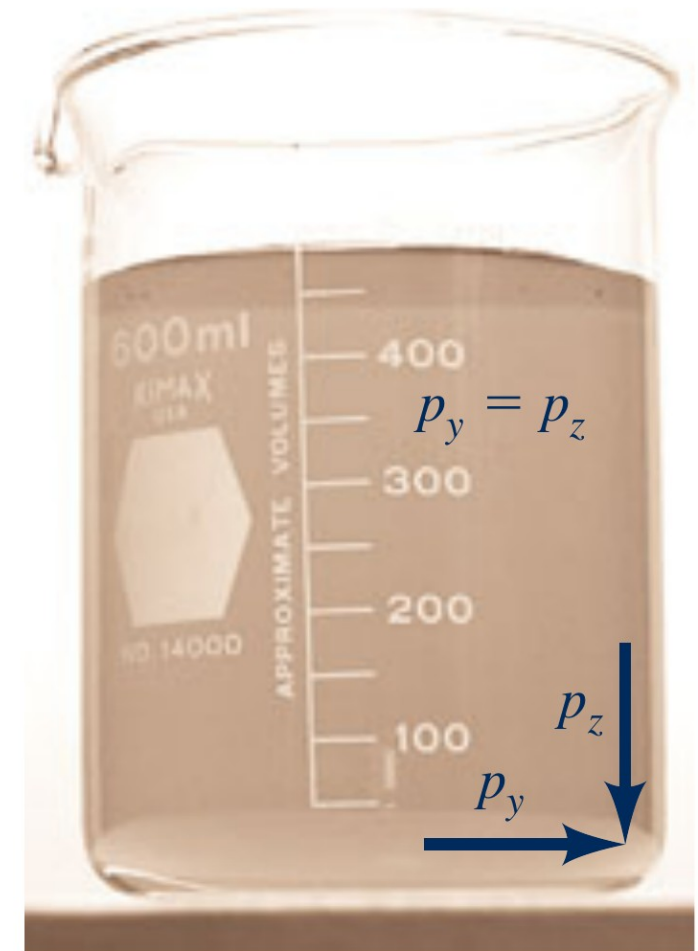
This is Pascal's law, after Blaise Pascal (1623–1662), who made important contributions to hydrostatics.

$$p_s = p_y = p_z$$

The condition attached to it is that there be no shearing stresses.

When adjacent particles move relative to one another, the normal stress at a point is not the same in every direction, and the pressure is defined as the average of three mutually perpendicular normal stresses.

➡ **The pressure at a point in a fluid at rest is independent of direction**



*At the junction of the side and the bottom of the beaker, the pressure is the same on both.*

# Basic equation for the pressure field

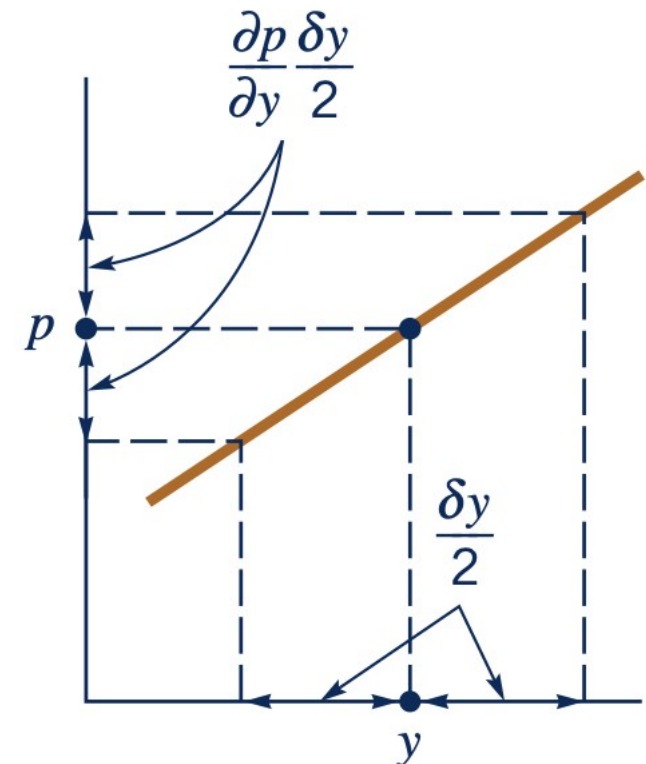
The pressure at a point is one number. **How does that number vary from point to point?** Take a small rectangular element of fluid. Two kinds of force act on it:

- **surface forces** due to the pressure on its six faces
- **a body force** equal to the weight of the element

With  $p$  the pressure at the centre, the average pressure on each face is written as a Taylor expansion about the centre, keeping only the first term. In the  $y$  direction:

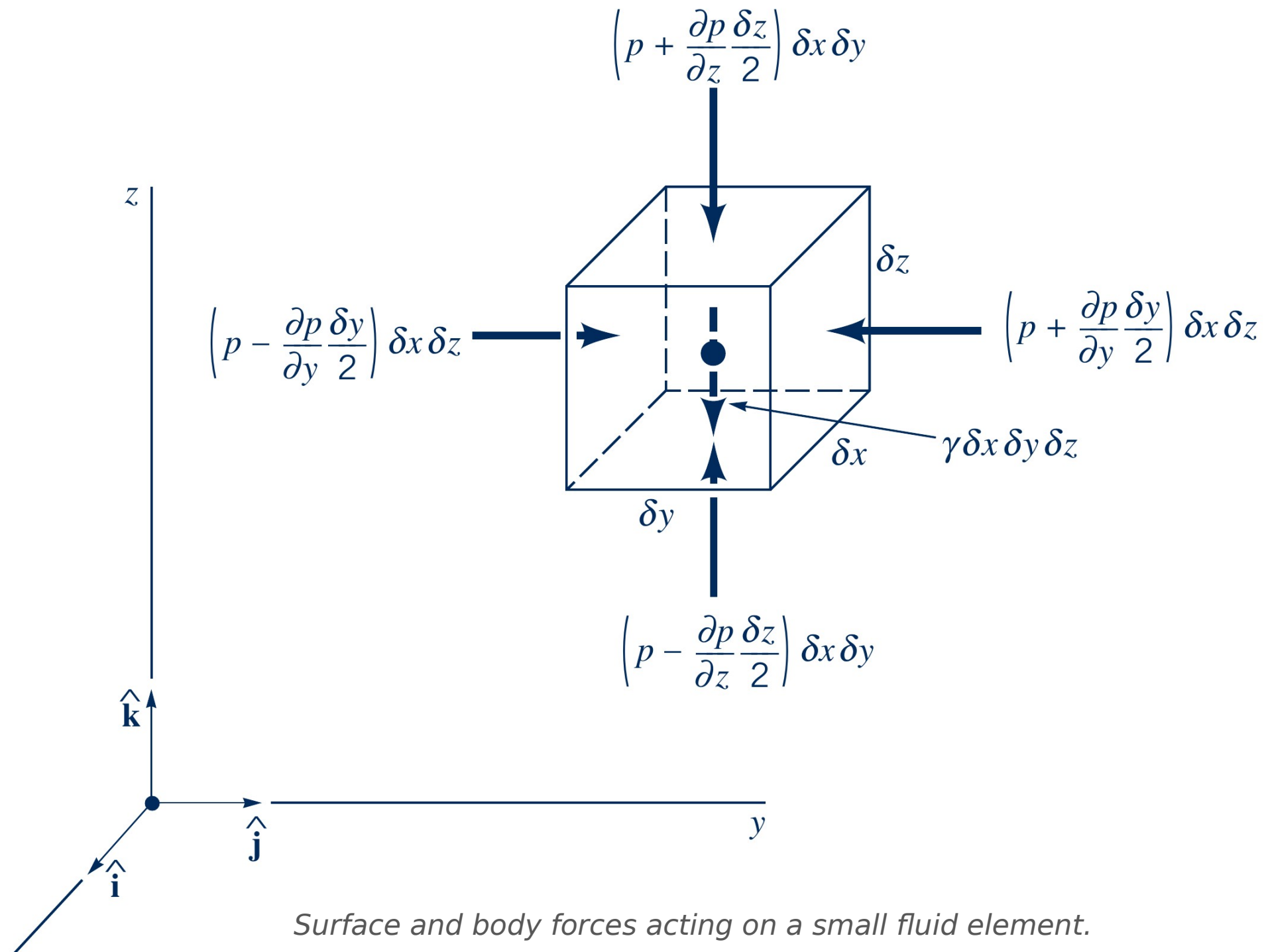
$$\begin{aligned}\delta F_y &= \left( p - \frac{\partial p}{\partial y} \frac{\delta y}{2} \right) \delta x \delta z - \left( p + \frac{\partial p}{\partial y} \frac{\delta y}{2} \right) \delta x \delta z \\ &= -\frac{\partial p}{\partial y} \delta x \delta y \delta z\end{aligned}$$

The pressure may vary across a fluid particle.



# Basic equation for the pressure field

- **surface forces** due to the pressure on its six faces
- **a body force** equal to the weight of the element



*Surface and body forces acting on a small fluid element.*

# The pressures on the faces: a Taylor expansion

The pressure itself cancels; only its derivative survives.

$$\begin{aligned}\delta F_y &= \left( p - \frac{\partial p}{\partial y} \frac{\delta y}{2} \right) \delta x \delta z - \left( p + \frac{\partial p}{\partial y} \frac{\delta y}{2} \right) \delta x \delta z \\ &= -\frac{\partial p}{\partial y} \delta x \delta y \delta z\end{aligned}$$

The same argument in x and z gives:

$$\delta F_x = -\frac{\partial p}{\partial x} \delta x \delta y \delta z, \quad \delta F_z = -\frac{\partial p}{\partial z} \delta x \delta y \delta z$$

In vectorial form:

$$\delta \mathbf{F}_s = \delta F_x \hat{i} + \delta F_y \hat{j} + \delta F_z \hat{k}$$

# The resultant surface force

Collecting the 3 components, the resultant surface force on the element is

$$\begin{aligned}\delta \mathbf{F}_s &= - \left( \frac{\partial p}{\partial x} \hat{i} + \frac{\partial p}{\partial y} \hat{j} + \frac{\partial p}{\partial z} \hat{k} \right) \delta x \delta y \delta z \\ &= -\nabla p \delta x \delta y \delta z\end{aligned}$$

The group in parentheses is the pressure gradient:

$$\frac{\partial p}{\partial x} \hat{i} + \frac{\partial p}{\partial y} \hat{j} + \frac{\partial p}{\partial z} \hat{k} = \nabla p$$

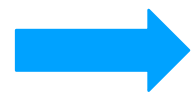
# Pressure gradient and the force per unit volume

$\nabla$  is the gradient, or del, operator:

$$\nabla( ) = \frac{\partial( )}{\partial x} \hat{i} + \frac{\partial( )}{\partial y} \hat{j} + \frac{\partial( )}{\partial z} \hat{k}$$

so the resultant surface force per unit volume is

$$\frac{\delta \mathbf{F}_s}{\delta x \delta y \delta z} = -\nabla p$$



**The surface force acting on a small fluid element depends only on the pressure gradient if there are no shearing stresses present.**

This is why the absolute level of the pressure never enters a hydrostatic force calculation: only differences matter, because only the gradient appears.

# The general equation of motion

The weight of the element acts downward, in the negative  $z$  direction:

$$-\delta\mathcal{W}\hat{k} = -\gamma \delta x \delta y \delta z \hat{k}$$

Newton's second law applied to the element, with mass  $\delta m = \rho \delta x \delta y \delta z$ :

$$\sum \delta \mathbf{F} = \delta \mathbf{F}_s - \delta \mathcal{W} \hat{k} = \delta m \mathbf{a}, \quad \delta m = \rho \delta x \delta y \delta z$$

Substituting the two force terms:

$$-\nabla p \delta x \delta y \delta z - \gamma \delta x \delta y \delta z \hat{k} = \rho \delta x \delta y \delta z \mathbf{a}$$

and dividing through by the volume of the element:

$$-\nabla p - \gamma \hat{k} = \rho \mathbf{a}$$

A fluid at rest is the special case  $\mathbf{a} = 0$ .



# Pressure variation in a fluid at rest

For a fluid at rest  $\mathbf{a} = \mathbf{0}$ , and the general equation reduces to

$$\nabla p + \gamma \hat{k} = 0$$

or, in component form,

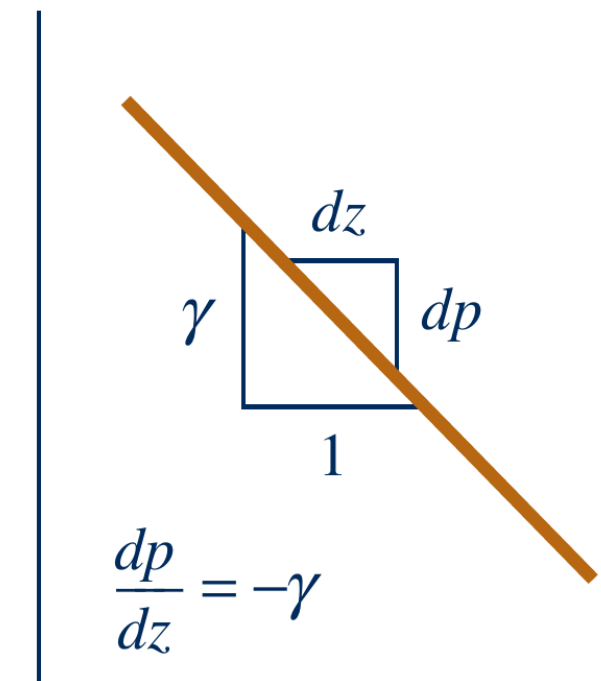
$$\frac{\partial p}{\partial x} = 0, \quad \frac{\partial p}{\partial y} = 0, \quad \frac{\partial p}{\partial z} = -\gamma$$

**The pressure does not depend on x or y:**

move anywhere within a horizontal plane and the pressure is unchanged.

Since p depends on z alone, the partial derivative becomes an ordinary one:

$$\frac{dp}{dz} = -\gamma$$



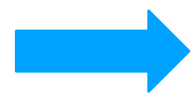
*The gradient is negative: pressure falls as we move up.*



# Fundamental equation for fluids at rest

$$\frac{dp}{dz} = -\gamma$$

- **The gradient is negative:** pressure decreases as we move upward in a fluid at rest.
- **There is no requirement that  $\gamma$  be constant:** the equation is valid for liquids, and for gases whose specific weight varies with elevation.
- **To integrate it we must first stipulate how  $\gamma$  varies with  $z$ ,** and that is what separates liquids from gases
- **It applies only at rest:** if the fluid is flowing, the pressure varies with  $x$ ,  $y$  and  $z$  in a way this equation does not describe

 **The vertical pressure gradient depends only on the local specific weight**

# Incompressible fluids

The specific weight is the product of density and gravity, so a change in  $\gamma$  comes from a change in  $\rho$  or in  $g$ .

$$\gamma = \rho g$$

For most purposes  $g$  is constant, so our main concern is the density.

**A fluid of constant density is called an incompressible fluid.** For liquids the variation is negligible even over large vertical distances, so the equation integrates directly:

$$\frac{dp}{dz} = -\gamma \quad \int_{p_1}^{p_2} dp = -\gamma \int_{z_1}^{z_2} dz$$

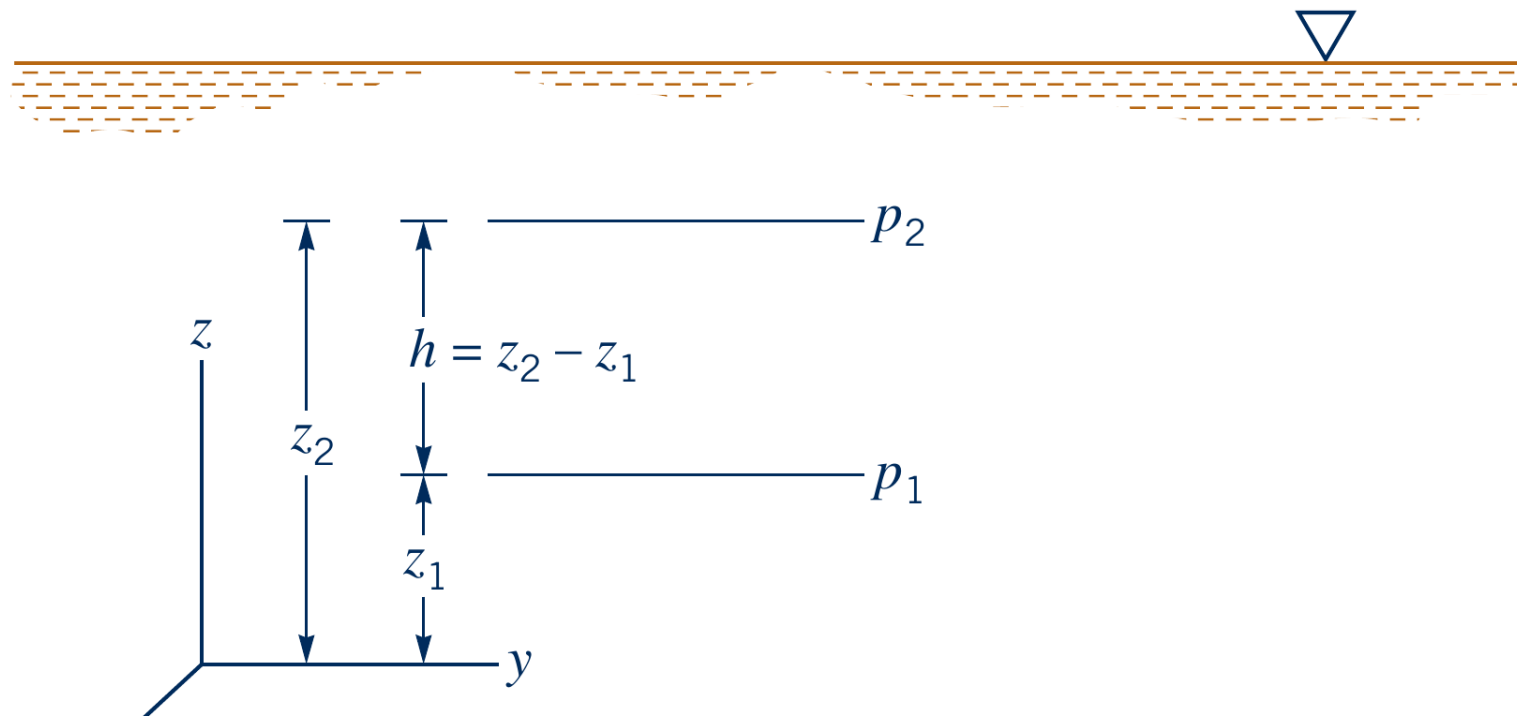
$$p_2 - p_1 = -\gamma (z_2 - z_1)$$

which is usually written with  $p_1$  the pressure at the lower elevation:

$$p_1 - p_2 = \gamma (z_2 - z_1)$$

# The hydrostatic pressure distribution

Notation for pressure variation in a fluid at rest with a free surface. Free surface (pressure =  $p_0$ )



With  $h = z_2 - z_1$  the depth measured downward from the level of  $p_2$ :

$$p_1 - p_2 = \gamma h$$

$$p_1 = \gamma h + p_2$$

Taking the free surface as the reference, where the pressure is  $p_0$ :

$$p = \gamma h + p_0$$

The pressure must increase with depth in order to hold up the weight of the fluid above it.

**➡ Pressure varies linearly with depth: the hydrostatic distribution**

# Pressure head ( $h$ )

The pressure difference between two points can be quoted as the height of a column of a stated fluid:

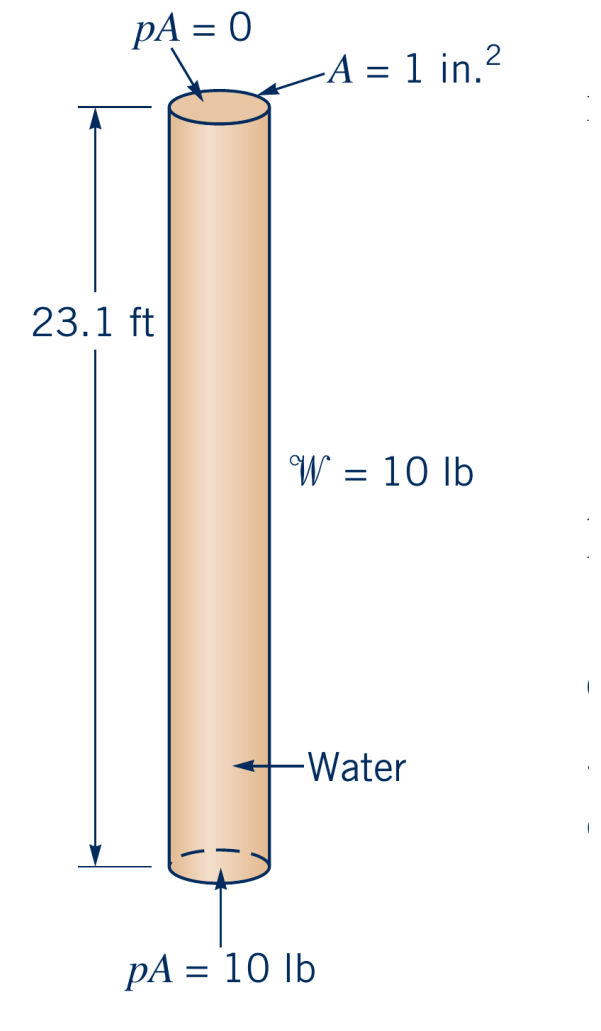
$$h = \frac{p_1 - p_2}{\gamma}$$

$h$  is the **pressure head**, interpreted as the height of a column of fluid of specific weight  $\gamma$  needed to produce that difference.

The fluid must always be named, because  $\gamma$  appears in the definition:

$$\Delta p = 69 \text{ kPa} : \quad h = 7.04 \text{ m of water} \\ = 518 \text{ mm of Hg}$$

This is why atmospheric pressure is quoted in millimetres of mercury, and pump performance in metres of water.



*A 23.1 ft column of water on 1 in<sup>2</sup> weighs 10 lb, that is, 10 psi.*

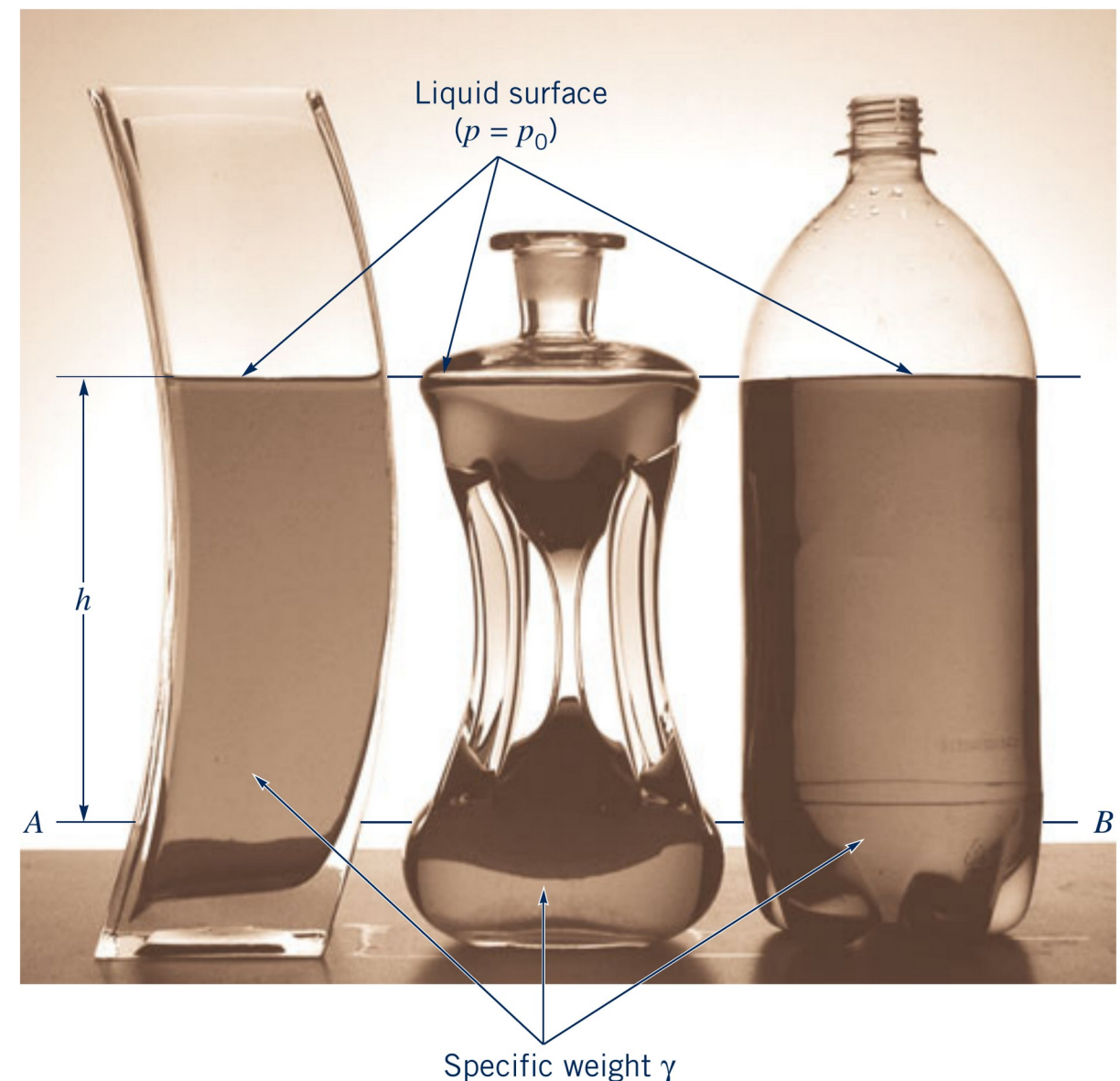
# The container shape does not matter

In a homogeneous, incompressible fluid at rest, the pressure at a point depends only on:

- the depth  $h$  below the free surface
- the surface pressure  $p_0$
- the specific weight  $\gamma$  of the liquid

It is not influenced by the size or the shape of the container.

Along the line AB the pressure is the same in all three vessels, however irregular they are.



# Pressure on a moving body is a different problem



*Pressure distribution on a car. © Stanford University*

The hydrostatic distribution holds only at rest. Once the fluid flows, the pressure varies with  $x$ ,  $y$  and  $z$ , as on the surface of a car driven along a road.

# Pressure on a moving body is a different problem

The pressure distribution on an object is entirely different when the object is stationary than when it is moving through a fluid.



*Pressure distribution on a car.*

For a stationary car (i.e., a hydrostatic condition) the pressure is the same on all surfaces of the car. This assumes that the pressure difference due to the difference in elevation from the top to the bottom of the car is negligible.

For a car moving through the air (i.e., a dynamic condition), the pressure distribution on the car's surface is very complex.

Shown here is a model car in a smoke tunnel. The resulting flow structure is quite complicated.

The pressure distribution on the car is highly non-uniform, with the highest pressure in the area of the front bumper and the lowest pressure in the irregular, turbulent wake region behind the car.



# How large is atmospheric pressure?

The sea level atmosphere exerts about **101 kPa of pressure** on a surface. We do not easily observe this pressure on everyday objects because the pressure acting on a typical body is the same on the inside and outside of it.



*A demonstration of atmospheric pressure.*



# How large is atmospheric pressure?

The effect of this sea level pressure becomes clear when the air inside the can is heated. The top is open, so hot air can flow out of the can and decrease the air inside.



*A demonstration of atmospheric pressure.*

The  $T$  of air inside the can increases while the  $p$  remains constant at atmospheric  $p$ . An increase in air  $T$  at constant  $p$  causes a decrease of the air  $\rho$  within the can that corresponds to the mass lost by the hot air escaping from the can.

The heated can is sealed and placed in a cold water bath. Initially, the hot air in the can is at atmospheric  $p$ . As the gas within the can cools and the  $T$  decreases at constant  $\rho$ , the  $p$  would drop below the atmospheric  $p$  outside.

The force created by the pressure imbalance is enough to crush the can. When the can collapses the volume within the can decreases, and the pressure increases.

101 kPa acting on every square metre is a force of about 10 tonnes. We do not notice it because it acts on both sides of every surface — until one side is evacuated.

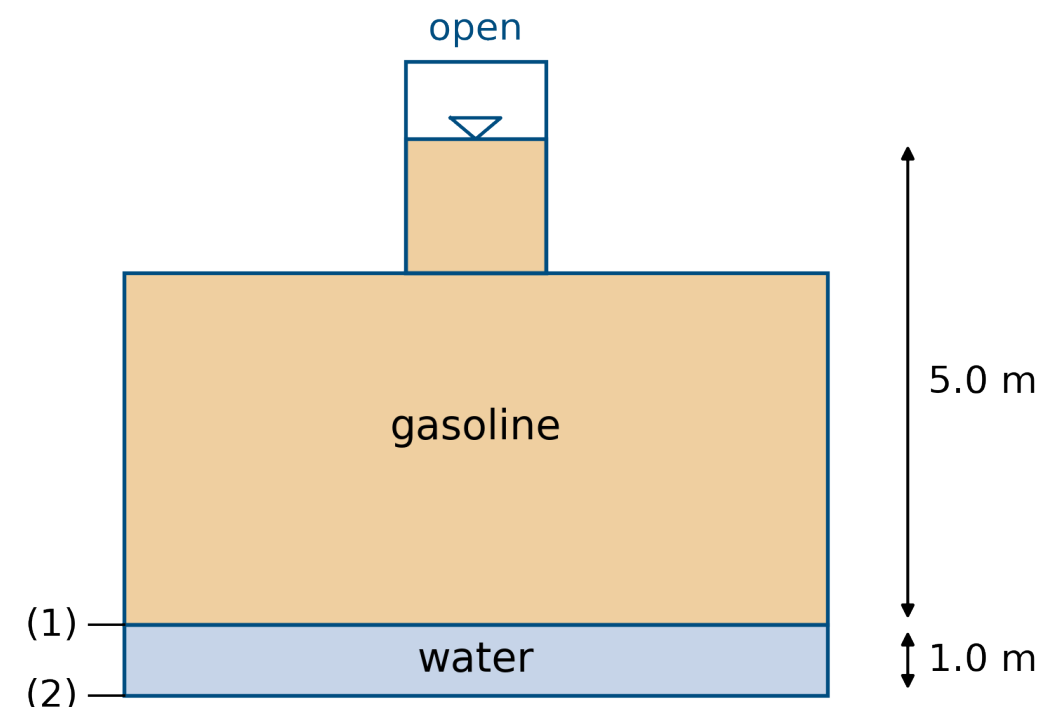
## Example — pressure in a leaking storage tank

Water has leaked into a buried gasoline storage tank and settled at the bottom. The gasoline layer is 5.0 m deep and the water layer 1.0 m deep, and the specific gravity of the gasoline is  $SG = 0.68$ . Find the gage pressure at the gasoline–water interface (1) and at the bottom of the tank (2), and express the second as a head of water.

$$\begin{aligned} p_1 &= SG \gamma_{\text{H}_2\text{O}} h \\ &= (0.68)(9.80 \text{ kN/m}^3)(5.0 \text{ m}) = 33.3 \text{ kPa} \end{aligned}$$

$$p_2 = \gamma_{\text{H}_2\text{O}} h_{\text{H}_2\text{O}} + p_1 = (9.80)(1.0) + 33.3 = 43.1 \text{ kPa}$$

$$\frac{p_2}{\gamma_{\text{H}_2\text{O}}} = \frac{43.1}{9.80} = 4.40 \text{ m of water}$$



# Transmitting pressure: the hydraulic jack

The requirement that pressures be equal at equal elevations is what makes hydraulic jacks, lifts, presses and aircraft controls work.

A piston at one end of a closed system filled with a liquid changes the pressure throughout it.

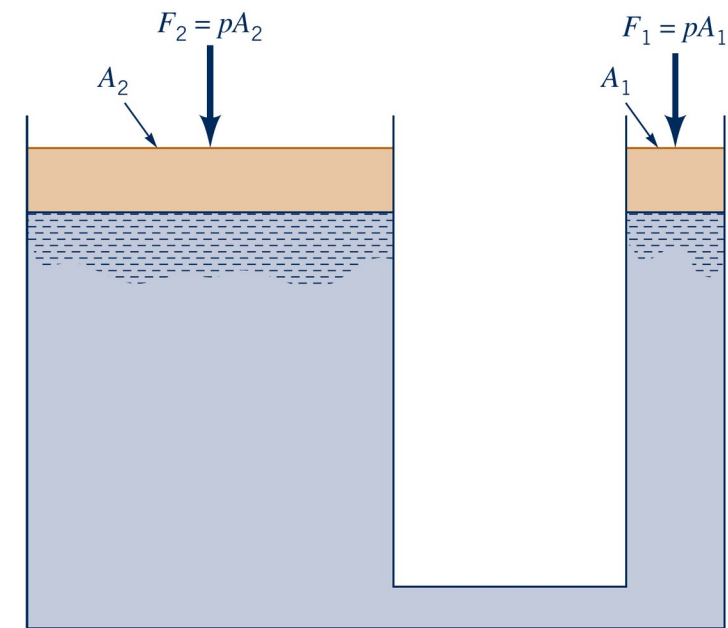
The same  $p$  acts on both pistons, and elevation differences are negligible for a device of this size, so:

$$F_2 = \frac{A_2}{A_1} F_1$$

$A_2$  can be made much larger than  $A_1$ , so a small applied force develops a large output force.



(a)



(b)

➡ **The principle behind many hydraulic devices**

# Compressible fluids

- Gases such as air, oxygen and nitrogen are compressible fluids, because their density changes significantly with changes in pressure and temperature

$$\frac{dp}{dz} = -\gamma$$

- This still applies at a point in a gas, but the variation of  $\gamma$  must be known before the equation can be integrated

- The specific weights of common gases are small compared with those of liquids:

$$\gamma_{\text{air}} = 11.99 \text{ N/m}^3, \quad \gamma_{\text{H}_2\text{O}} = 9.80 \times 10^3 \text{ N/m}^3 \quad (15.6^\circ\text{C, sea level})$$

$$\frac{\gamma_{\text{H}_2\text{O}}}{\gamma_{\text{air}}} \approx 800$$

Air is some 800 times lighter than water, so **the vertical pressure gradient in a gas is correspondingly small.**

$$\Delta p = \gamma \Delta z = (11.99 \text{ N/m}^3)(100 \text{ m}) = 1.2 \text{ kPa} \approx 1.2\% \text{ of } p_{\text{atm}}$$

 **For a gas in a tank or a pipe, elevation changes can be neglected**

# Compressible fluid

In general for compressible fluids (e.g. gases),  $\gamma$  cannot be taken outside the integral.  
Using the equation of state for an ideal gas:

$$\rho = \frac{p}{RT} \qquad \frac{dp}{dz} = -\frac{g p}{RT}$$

Separating the variables and integrating between two elevations:

$$\int_{p_1}^{p_2} \frac{dp}{p} = \ln \frac{p_2}{p_1} = -\frac{g}{R} \int_{z_1}^{z_2} \frac{dz}{T}$$

$g$  and  $R$  are taken as constant over the elevation change.

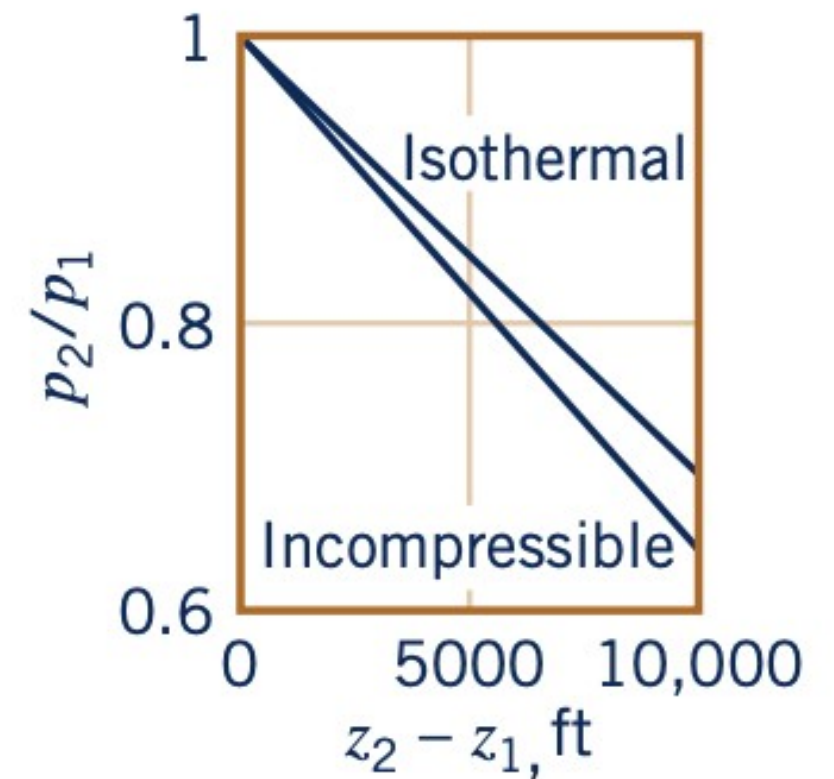
Before integrating the right-hand side, we must specify how  $T$  varies with  $z$ .

# An isothermal layer

Assuming a constant temperature  $T_0$  over the range  $z_1$  to  $z_2$ :

$$p_2 = p_1 \exp \left[ - \frac{g (z_2 - z_1)}{R T_0} \right]$$

- The pressure falls exponentially with height, not linearly
- The scale height  $RT_0/g$  is about 8.4 km for air at
- Even over a 3 km change, the isothermal and the incompressible results differ very little



**➡ If  $\gamma$  varies significantly, pressure no longer varies linearly with depth**



# Example — pressure at the top of a tall building

The Burj Khalifa in the United Arab Emirates is 828 m (2717 ft) tall. Estimate the ratio of the pressure at the top to the pressure at the base, (a) treating the air as isothermal at 15 °C, and (b) treating it as incompressible with  $\gamma = 12.014$  N/m<sup>3</sup> at 101.33 kPa (abs).

$$\frac{p_2}{p_1} = \exp \left[ -\frac{(9.81)(828)}{(286.9)(288)} \right] = 0.906$$

$$\begin{aligned} \frac{p_2}{p_1} &= 1 - \frac{\gamma (z_2 - z_1)}{p_1} \\ &= 1 - \frac{(12.014)(828)}{101,330} = 0.902 \end{aligned}$$

➡ **The pressure falls by about 10% over the tallest building on Earth**

The **two models agree to within half a percent**, which justifies treating a gas as incompressible whenever the height involved is small.



■ **Figure E2.2** (Figure courtesy of Emaar Properties, Dubai, UAE.)

# The standard atmosphere

The **standard atmosphere** is an idealised representation of mean conditions in the Earth's atmosphere.

Measurements of pressure against altitude for the specific conditions of interest are rarely available, so a standard atmosphere is used to design and compare aircraft, missiles and spacecraft.

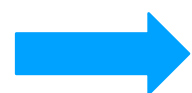
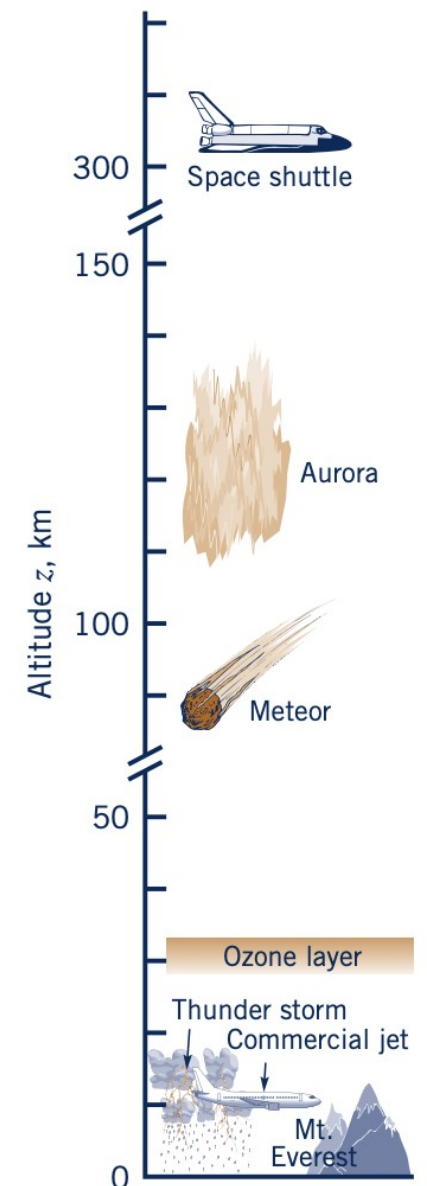
The standard atmosphere is a representation of middle-latitude, year-round mean conditions of the Earth's atmosphere.

■ **Table 2.1**

**Properties of U.S. Standard Atmosphere at Sea Level\***

| Property                  | SI Units                                    | BG Units                                                              |
|---------------------------|---------------------------------------------|-----------------------------------------------------------------------|
| Temperature, $T$          | 288.15 K (15 °C)                            | 518.67 °R (59.00 °F)                                                  |
| Pressure, $p$             | 101.33 kPa (abs)                            | 2116.2 lb/ft <sup>2</sup> (abs)<br>[14.696 lb/in. <sup>2</sup> (abs)] |
| Density, $\rho$           | 1.225 kg/m <sup>3</sup>                     | 0.002377 slugs/ft <sup>3</sup>                                        |
| Specific weight, $\gamma$ | 12.014 N/m <sup>3</sup>                     | 0.07647 lb/ft <sup>3</sup>                                            |
| Viscosity, $\mu$          | $1.789 \times 10^{-5}$ N · s/m <sup>2</sup> | $3.737 \times 10^{-7}$ lb · s/ft <sup>2</sup>                         |

\*Acceleration of gravity at sea level =  $9.807 \text{ m/s}^2 = 32.174 \text{ ft/s}^2$ .



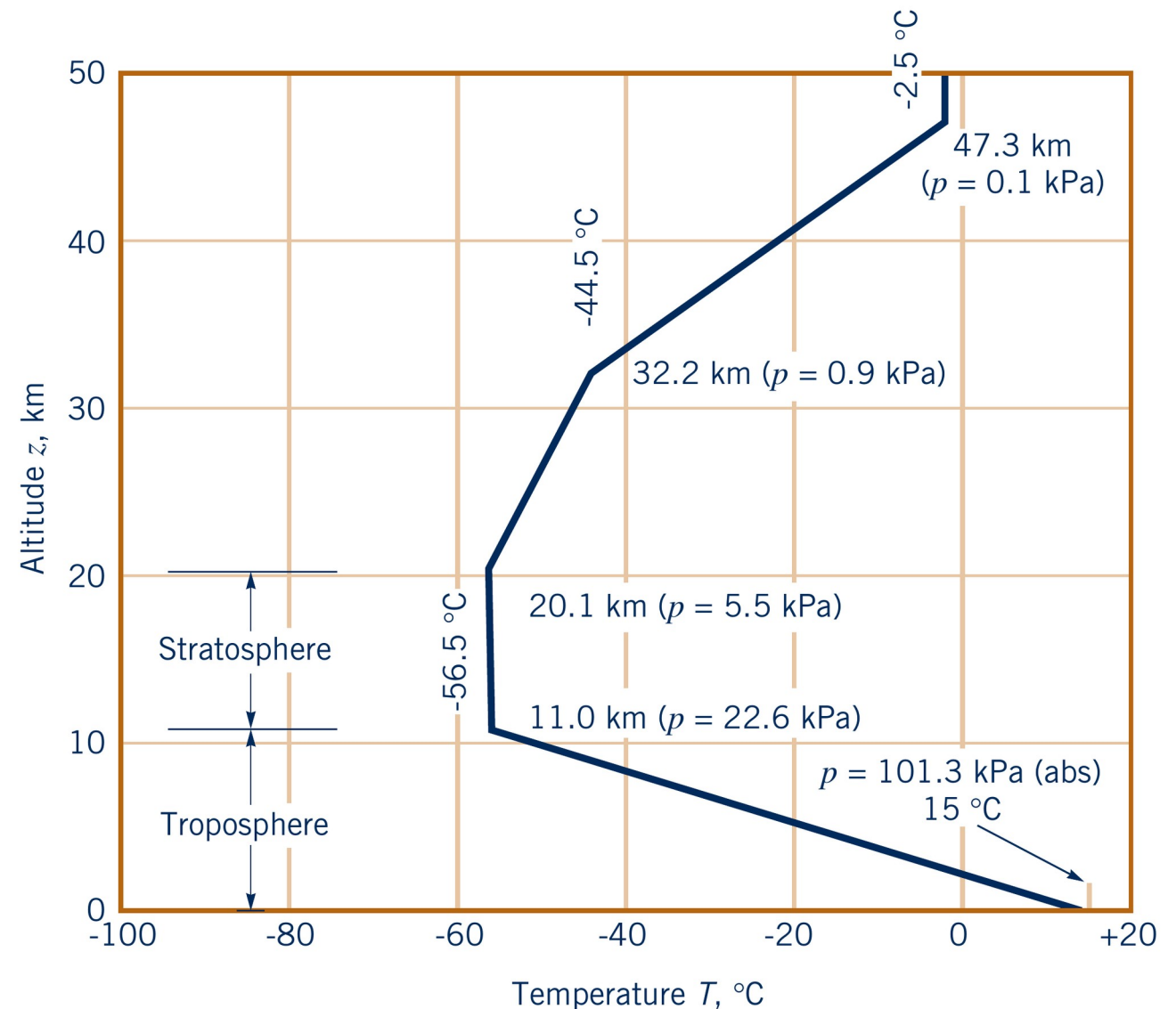
**An idealised representation of mean atmospheric conditions**



# Temperature profile of the standard atmosphere

The temperature falls with altitude in the region nearest the surface, the **troposphere**; becomes essentially constant through the **stratosphere**; and rises again in the layer above. Earth's surface is the main source of heating for the troposphere.

Ozone absorbs ultraviolet radiation from the Sun, heating the upper stratosphere.



*Variation of temperature with altitude in the U.S. standard atmosphere.*

Since the profile is a series of straight segments, the eq. below can be integrated segment by segment to give the pressure at any altitude.

$$\int_{p_1}^{p_2} \frac{dp}{p} = \ln \frac{p_2}{p_1} = -\frac{g}{R} \int_{z_1}^{z_2} \frac{dz}{T}$$

# Temperature profile of the standard atmosphere

This table gives the tabulated temperature, gravity, pressure, density and viscosity.

■ **Table C.2**

**Properties of the U.S. Standard Atmosphere (SI Units)<sup>a</sup>**

| Altitude<br>(m) | Temperature<br>(°C) | Acceleration<br>of Gravity,<br>$g$ (m/s <sup>2</sup> ) | Pressure, $p$<br>[N/m <sup>2</sup> (abs)] | Density,<br>$\rho$<br>(kg/m <sup>3</sup> ) | Dynamic<br>Viscosity,<br>$\mu$<br>(N·s/m <sup>2</sup> ) |
|-----------------|---------------------|--------------------------------------------------------|-------------------------------------------|--------------------------------------------|---------------------------------------------------------|
| −1,000          | 21.50               | 9.810                                                  | 1.139 E + 5                               | 1.347 E + 0                                | 1.821 E − 5                                             |
| 0               | 15.00               | 9.807                                                  | 1.013 E + 5                               | 1.225 E + 0                                | 1.789 E − 5                                             |
| 1,000           | 8.50                | 9.804                                                  | 8.988 E + 4                               | 1.112 E + 0                                | 1.758 E − 5                                             |
| 2,000           | 2.00                | 9.801                                                  | 7.950 E + 4                               | 1.007 E + 0                                | 1.726 E − 5                                             |
| 3,000           | −4.49               | 9.797                                                  | 7.012 E + 4                               | 9.093 E − 1                                | 1.694 E − 5                                             |
| 4,000           | −10.98              | 9.794                                                  | 6.166 E + 4                               | 8.194 E − 1                                | 1.661 E − 5                                             |
| 5,000           | −17.47              | 9.791                                                  | 5.405 E + 4                               | 7.364 E − 1                                | 1.628 E − 5                                             |
| 6,000           | −23.96              | 9.788                                                  | 4.722 E + 4                               | 6.601 E − 1                                | 1.595 E − 5                                             |
| 7,000           | −30.45              | 9.785                                                  | 4.111 E + 4                               | 5.900 E − 1                                | 1.561 E − 5                                             |
| 8,000           | −36.94              | 9.782                                                  | 3.565 E + 4                               | 5.258 E − 1                                | 1.527 E − 5                                             |
| 9,000           | −43.42              | 9.779                                                  | 3.080 E + 4                               | 4.671 E − 1                                | 1.493 E − 5                                             |
| 10,000          | −49.90              | 9.776                                                  | 2.650 E + 4                               | 4.135 E − 1                                | 1.458 E − 5                                             |
| 15,000          | −56.50              | 9.761                                                  | 1.211 E + 4                               | 1.948 E − 1                                | 1.422 E − 5                                             |
| 20,000          | −56.50              | 9.745                                                  | 5.529 E + 3                               | 8.891 E − 2                                | 1.422 E − 5                                             |
| 25,000          | −51.60              | 9.730                                                  | 2.549 E + 3                               | 4.008 E − 2                                | 1.448 E − 5                                             |
| 30,000          | −46.64              | 9.715                                                  | 1.197 E + 3                               | 1.841 E − 2                                | 1.475 E − 5                                             |
| 40,000          | −22.80              | 9.684                                                  | 2.871 E + 2                               | 3.996 E − 3                                | 1.601 E − 5                                             |
| 50,000          | −2.50               | 9.654                                                  | 7.978 E + 1                               | 1.027 E − 3                                | 1.704 E − 5                                             |
| 60,000          | −26.13              | 9.624                                                  | 2.196 E + 1                               | 3.097 E − 4                                | 1.584 E − 5                                             |
| 70,000          | −53.57              | 9.594                                                  | 5.221 E + 0                               | 8.283 E − 5                                | 1.438 E − 5                                             |
| 80,000          | −74.51              | 9.564                                                  | 1.052 E + 0                               | 1.846 E − 5                                | 1.321 E − 5                                             |

<sup>a</sup>Data abridged from *U.S. Standard Atmosphere*, 1976, U.S. Government Printing Office, Washington, D.C.

# Pressure in the troposphere

In the troposphere, which reaches about 11 km, the temperature falls linearly with altitude:

$$T = T_a - \beta z$$

where  $T_a$  is the sea-level temperature and  $\beta$  is the lapse rate. Substituting into the integral and carrying it out:

$$p = p_a \left( 1 - \frac{\beta z}{T_a} \right)^{g/R\beta}$$

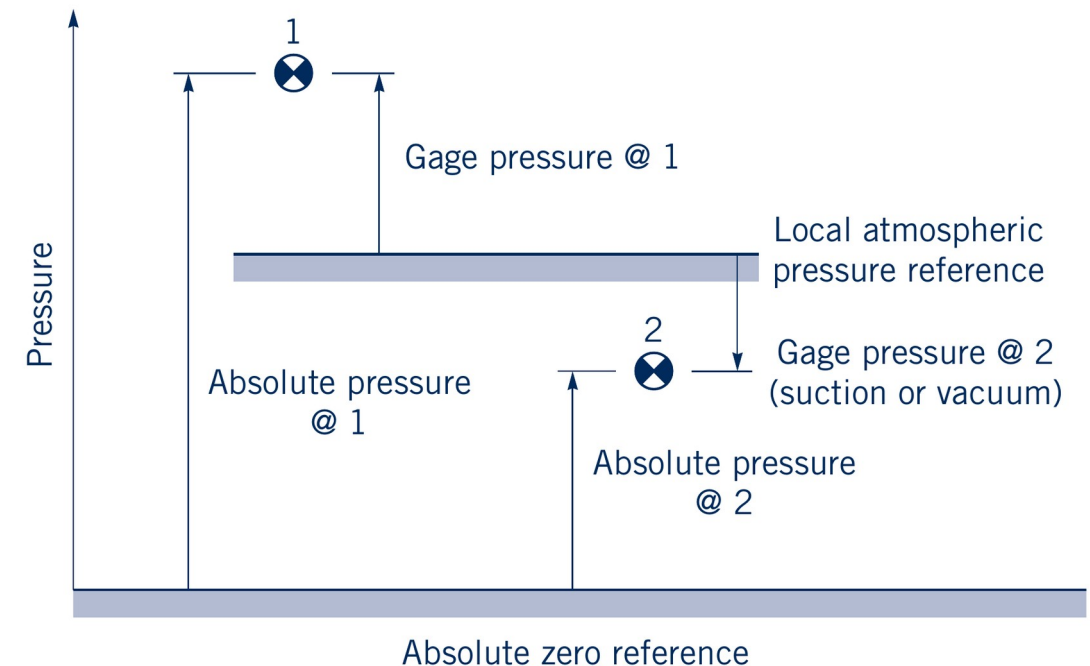
$$\beta = 0.00650 \text{ K/m}, \quad R = 286.9 \text{ J/(kg} \cdot \text{K)}$$

$$z = 11 \text{ km} : \quad T = -56.5^\circ\text{C}, \quad p \approx 23 \text{ kPa (abs)}$$

Modern jetliners cruise at about that altitude, where the outside pressure is roughly a fifth of the sea-level value.

# Absolute and gage pressure

- **Absolute pressure** is measured relative to a perfect vacuum, and is always positive
- **Gage pressure** is measured relative to the local atmospheric pressure, and can be positive or negative
- **A negative gage pressure** is called a suction or a vacuum pressure



$$p_{\text{abs}} = p_{\text{gage}} + p_{\text{atm}}$$

$$\begin{aligned} 10 \text{ kPa (abs)} &\equiv -91.3 \text{ kPa (gage)} \\ &\equiv 91.3 \text{ kPa vacuum} \end{aligned}$$

**Convention in the textbook and in this course:** a pressure is a gage pressure unless it is explicitly marked (abs).

# The mercury barometer

A tube closed at one end, filled with mercury and inverted into a mercury reservoir. The column settles where its weight, plus the force due to the vapour above it, balances the atmospheric pressure:

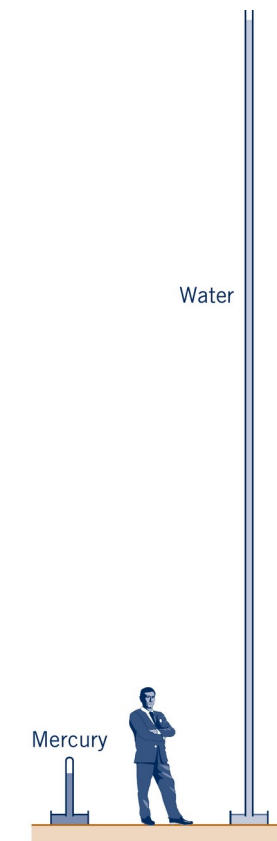
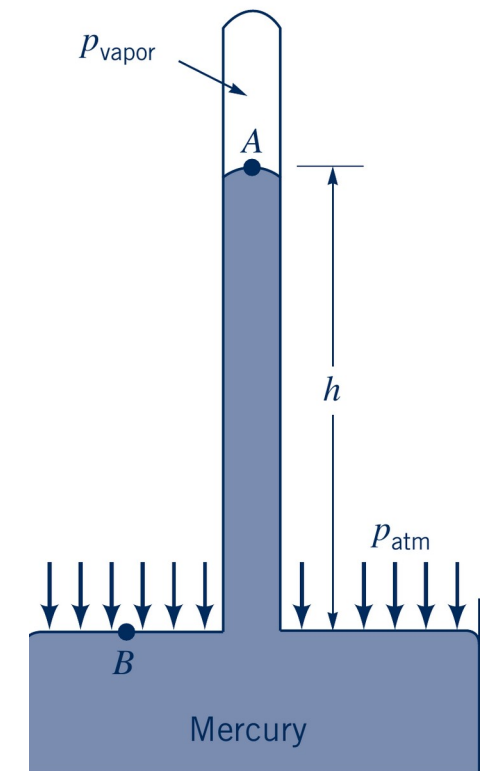
$$p_{\text{atm}} = \gamma h + p_{\text{vapor}} \simeq \gamma h$$

For mercury the vapour pressure is only 0.17 Pa (abs) at 20 °C, entirely negligible here.

$$h_{\text{Hg}} = 760 \text{ mm} \iff h_{\text{H}_2\text{O}} = 10.3 \text{ m}$$

A water barometer would need a tube about 10 m tall.

The barometer was invented by Evangelista Torricelli, about 1644.



*Mercury against water, to scale.*

## Example — pressure at the bottom of a mountain lake

A mountain lake has an average temperature of 10 °C and a maximum depth of 40 m. The barometric pressure at the surface is 598 mm Hg. Determine the absolute pressure, in pascals, at the deepest point.

The surface pressure is the barometric reading, converted with the specific weight of mercury:

$$p_0 = (0.598 \text{ m})(133 \text{ kN/m}^3) = 79.5 \text{ kPa}$$

and with  $\gamma = 9.804 \text{ kN/m}^3$  for water at 10 °C:

$$p = (9.804 \text{ kN/m}^3)(40 \text{ m}) + 79.5 = 472 \text{ kPa (abs)}$$

# Lecture summary

In a fluid with no shearing stresses the pressure at a point is the same in every direction, and its variation from point to point is set entirely by the weight of the fluid. Everything below is an integration of  $dp/dz = -\gamma$  under a stated assumption about  $\gamma$ , or a definition built on it.

$$p_s = p_y = p_z$$

pressure at a point

$$-\nabla p - \gamma \hat{k} = \rho \mathbf{a}$$

no shearing stresses

$$\frac{dp}{dz} = -\gamma$$

fluid at rest

$$p = \gamma h + p_0$$

hydrostatic distribution

$$h = \frac{\Delta p}{\gamma}$$

pressure head

$$F_2 = \frac{A_2}{A_1} F_1$$

hydraulic transmission

$$p_2 = p_1 e^{-g(z_2 - z_1)/RT_0}$$

isothermal gas layer

$$p = p_a \left(1 - \frac{\beta z}{T_a}\right)^{g/R\beta}$$

troposphere

$$p_{\text{abs}} = p_{\text{gage}} + p_{\text{atm}}$$

absolute and gage pressure

$$p_{\text{atm}} \simeq \gamma h$$

the barometer